Final Feature List

Detection and Attribution of AI-Generated and Human Written Phishing Emails from Legitimate Emails

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Overview

The pipeline extracts **20 features** from each email, split across two parallel tracks by feature type. **14 continuous features** are screened with Kruskal-Wallis H (3-group: Legitimate vs. Human-Phishing vs. AI-Phishing) and, on the phishing-only subset, Mann-Whitney U (2-group attribution: Human-Phishing vs. AI-Phishing). **6 categorical/flag features** are screened with Chi-Square (3-group, and a 2-group attribution variant). Every raw p-value from both tracks is corrected jointly with Benjamini-Hochberg FDR, and every feature must also clear an effect-size floor (Cohen's $d \geq 0.20$ for continuous, Cramér's $V \geq 0.10$ for categorical) before it reaches the Naive Bayes classification stage. This document lists all 20 candidate features; which ones actually survive both gates on the full training set is determined by running the pipeline itself, not decided in advance.

1 Continuous Features (14) — Kruskal-Wallis H / Mann-Whitney U

#	Feature	Category	Description / lane
1	Average token log-probability	Text Complexity	How predictable each word is under a language model; AI text tends to be more predictable (higher log-prob). Main track.
2	Sentence perplexity variance (burstiness)	Text Complexity	Variance of per-sentence perplexity; human writing is "bursty" (varies a lot), AI text is uniform. Main track.
3	Standard deviation of sentence length	Structure & Syntax	Spread of sentence lengths within an email. Main track.
4	Maximum parse-tree depth	Structure & Syntax	Deepest syntactic nesting found in the email. Main track.
5	Paragraph density	Structure & Syntax	Paragraph breaks per 100 words. Main track.

#	Feature	Category	Description / lane
6	Clause density	Structure & Syntax	Linking words ("and", "because", "which", ...) per sentence. Main track.
7	Orthographic error density	Language Mechanics	Spelling/typo flags per 100 words. Main track.
8	Punctuation distribution entropy	Language Mechanics	Shannon entropy over punctuation-mark frequencies. Main track.
9	Urgency-keyword density	Communication Style	Urgency words ("urgent", "immediately", "asap", ...) per 100 words. Main **and** attribution track.
10	Pronoun balance (1st:2nd person ratio)	Communication Style	Ratio of first-person to second-person pronoun counts. Main track.
11	Sentiment variance across sentences	Communication Style	Variance of per-sentence sentiment polarity. Main track.
12	Imperative verb density	Communication Style	Command verbs ("click", "verify", "submit", ...) per 100 words. Main **and** attribution track — the strongest attribution signal in the feature set.
13	Lexical diversity (type-token ratio)	Lexical Richness	Unique words ÷ total words. Main track.
14	Pronoun density (overall)	Lexical Richness	Share of all words that are any pronoun (distinct from #10's first:second-person ratio). Main track.

2 Categorical / Flag Features (6) — Chi-Square

These six were validated directly against the 4,500-row training set (see Section 3) and are the first features to actually populate the Chi-Square branch of the pipeline, which was previously defined in the architecture but had no features assigned to it.

#	Feature	Description
15	Generic greeting flag	"Dear Valued Customer / Dear User" style opening vs. a named greeting.
16	Subject RE:/FW: flag	Subject line starts with a reply/forward marker.
17	Attachment-mention flag	Email references an attached file.
18	Bracketed call-to-action flag	A bracketed action phrase, e.g. `[Click Here]`, `[Verify Now]`.
19	Sensitive-info-request flag	Email asks for a password, SSN, card number, CVV, or PIN.

#	Feature	Description
20	Threat/consequence-clause flag	Email threatens account suspension, closure, or restriction.

3 Validation on the training set

Each categorical feature was tested with a real χ^2 test of independence on the 4,500-row balanced training set (1,500 Legitimate / 1,500 Human-Phishing / 1,500 AI-Phishing), both as a 3-group test and restricted to the phishing-only subset for attribution. A feature is marked as clearing the pipeline's gate only where adjusted $p < 0.05$ **and** Cramér's $V \geq 0.10$.

	3-group (main)				2-group (attribution)			
Feature	χ^2	p	V	**Gate**	χ^2	p	V	**Gate**
Generic greeting	1759.7	≈ 0	0.625	✓	847.4	≈ 0	0.531	✓
Subject RE:/FW:	1192.0	≈ 0	0.515	✓	69.6	7.1×10^{-17}	0.152	✓*
Sensitive-info request	154.7	2.6×10^{-34}	0.185	✓	20.9	4.9×10^{-6}	0.083	**fails**
Attachment mention	147.6	8.9×10^{-33}	0.181	✓	53.5	2.5×10^{-13}	0.134	✓
Threat/consequence clause	146.2	1.8×10^{-32}	0.180	✓	19.1	1.3×10^{-5}	0.080	**fails**
Bracketed CTA	130.7	4.2×10^{-29}	0.170	✓	83.4	6.9×10^{-20}	0.167	✓

**Caveat*: in the training corpus, 0.0% of AI-Phishing subjects start with RE:/FW: (not merely rare — exactly zero across all 1,500 rows), most likely an artifact of the fixed generation pretexts used to build the AI-Phishing corpus rather than an inherent property of AI-generated phishing. The Legitimate-vs-Phishing separation this feature captures is still genuine; the Human-vs-AI split should not be over-claimed as general in the written-up results.

Result: 4 of 6 categorical features clear both gates for both the main and attribution tasks. Sensitive-info-request and threat/consequence-clause clear the main-task gate (they do help separate Legitimate from Phishing) but fall under the effect-size floor for attribution specifically — statistically significant due to sample size, but too small an effect to use for telling Human-Phishing and AI-Phishing apart.

4 Full combined list

1. Average token log-probability
2. Sentence perplexity variance (burstiness)
3. Standard deviation of sentence length
4. Maximum parse-tree depth
5. Paragraph density
6. Clause density
7. Orthographic error density
8. Punctuation distribution entropy
9. Urgency-keyword density

10. Pronoun balance (1st:2nd person ratio)
11. Sentiment variance across sentences
12. Imperative verb density
13. Lexical diversity (type-token ratio)
14. Pronoun density (overall)
15. Generic greeting flag
16. Subject RE:/FW: flag
17. Attachment-mention flag
18. Bracketed call-to-action flag
19. Sensitive-info-request flag
20. Threat/consequence-clause flag